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SEVENTH SEMESTER B. Tech EXAMINATION, DECEMBER–2024

PAVEMENT DESIGN

Time:3 Hours **Maximum Marks:50**

Instructions:
- **i.** PART - A (Question 1-5) answer any 4 questions.
- **ii.** PART - B (Question 6-8) answer any 2 questions.
- **iii.** PART - C (Question 9) is compulsory

PART–A 04X05=20

1. Calculate the design deflection of a single-layer flexible pavement, given the following parameters: contact pressure at the road surface due to wheel load is 2 MPa, the radius of the contact area is 0.2 m, and the modulus of elasticity for the subgrade is 25 MPa. **05**

2. Enlist any five key input parameters required for flexible pavement design using the IRC method. **05**

3. State the assumptions made in the three-layered elastic pavement system. **05**

4. The design thickness of a CC pavement is 25 cm and the spacing between longitudinal joints is 3.5 m. If the allowable tensile stress in concrete during the initial curing period is 0.8 kg/cm^2, determine the spacing between the contraction joints in order to prevent random cracking across the CC pavement. The unit weight of CC = 2400 kg/m^3 and the friction coefficient between the membrane supporting layer and bottom of CC slab = 1.2. **05**

5. Outline the steps involved in the design process for a concrete slab used in rigid pavement. **05**

PART - B 02X09=18

6. The design thickness of cement concrete pavement is 26 cm considering design axle load (98th percentile load) of 12000 kg on single axle and M40 concrete with characteristic compressive strength of 400 kg/cm^2. The radius of relative stiffness is found to be 62.2 cm. If the elastic modulus of dowel bar steel is 2 x 10^6 kg/cm^2. Modulus of dowel concrete interaction is 41500 kg/cm^3 and joint width is 1.8 cm. Design the dowel bars for 40% load transfer considering edge loading (as per IRC 58-2011). **09**

7. a. Analyze the differences between highway pavements and airfield pavements. **05**

b. A flexible pavement of thickness 48 cm is laid over a subgrade. A circular load of radius 16 cm with uniform contact pressure of 6.5 kg/cm^2 is applied. Assuming homogeneous elastic single layer, determine the deflection of the pavement surface under the center of the load. Assume the elastic modulus of the subgrade as well as pavement layer to be 900 kg/cm^2. The deflection Chart can be found below Figure. 7. **04**

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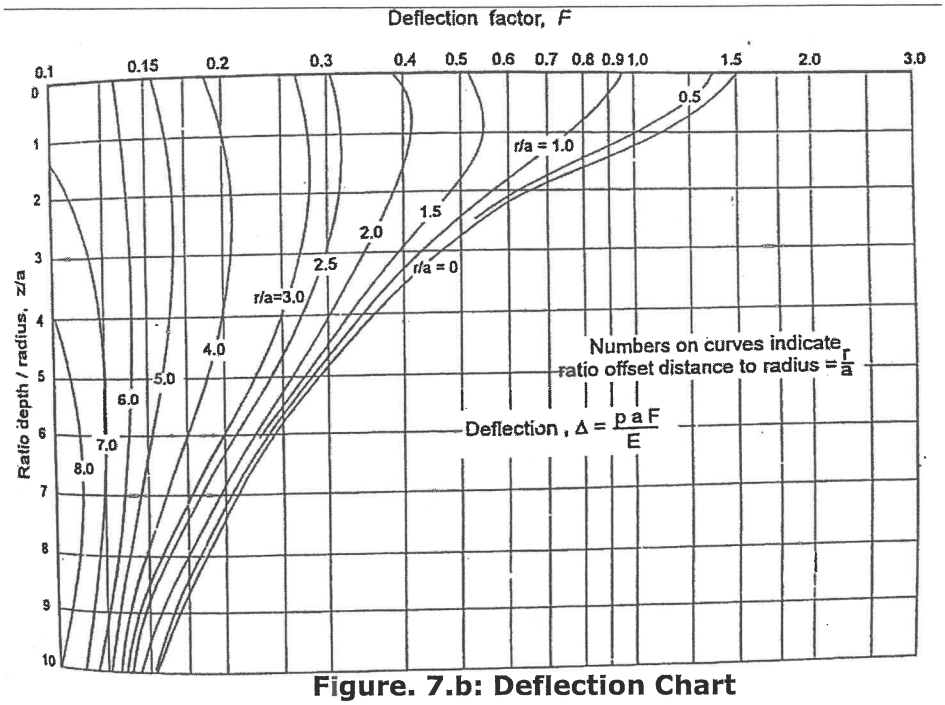

Figure. 7.b: Deflection Chart

8. **a.** A cement concrete slab has a length of 5 m with width 3.5 m and thickness 25 cm. The temperature differential between the top and bottom of concrete slab is 10° C. Calculate the maximum warping stress at the slab center. Assume E = 30 GPa; $\propto = 10 \times 10^{-6}/^{\circ}C$; $\mu = 1.5$. **06**

b. A concrete pavement is subjected to a wheel load stress of 400 kPa, a warping stress of 1.5 MPa and frictional stress of 5 kPa. Calculate the total combined stress. **03**

PART - C **01X12=12**

9. Soil sub-grade sample obtained from the site was undergone CBR test in laboratory which gives 2.5 mm penetration is 48.5 kg and 5.00 mm penetration is 75.2 kg. **12**

Following details for different pavement layers

- **i.** Compacted sandy soil 8% CBR
- **ii.** Poorly graded gravel 20% CBR
- **iii.** Well graded gravel 95% CBR
- **iv.** Minimum thickness of bituminous concrete surfacing is 5cm

The CBR chart can be found below figure. 9. The traffic survey revealed the present average daily traffic of commercial vehicles as 1500. The annual growth rate of traffic is 9%. The pavement construction is to be completed in 3 years after the last traffic count. Design the pavement as per IRC method.

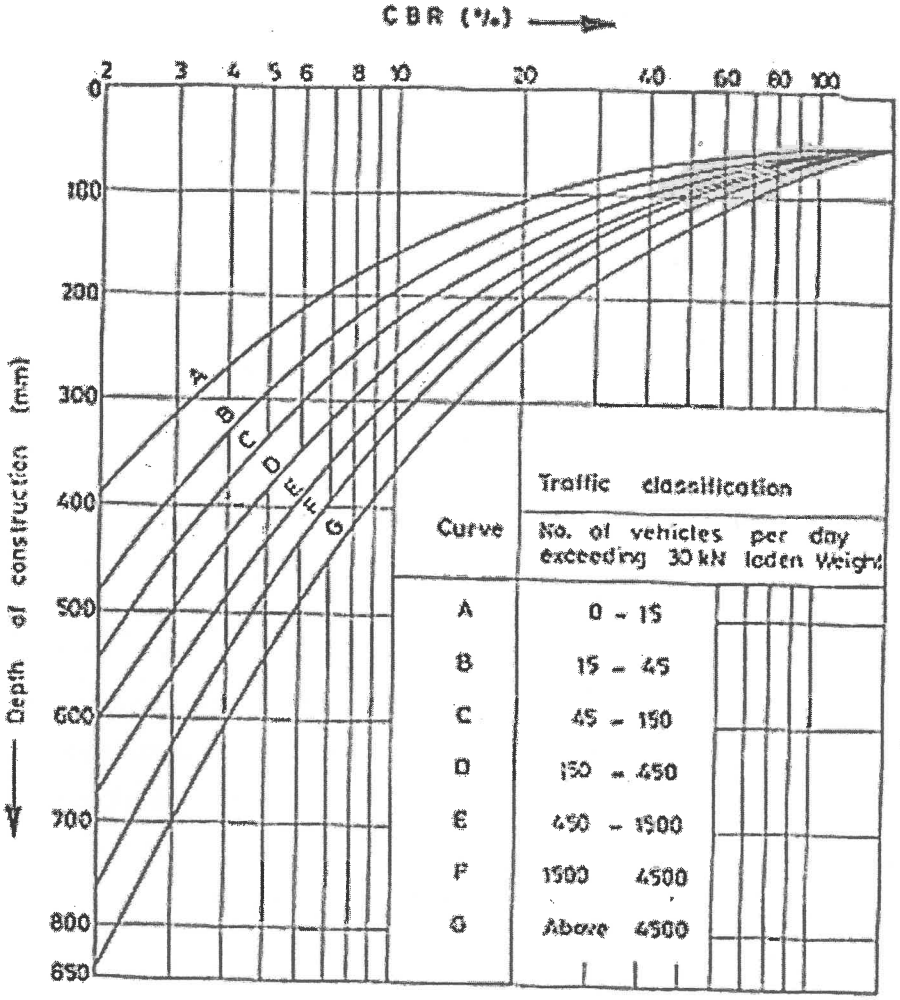

Figure. 9: CBR chart